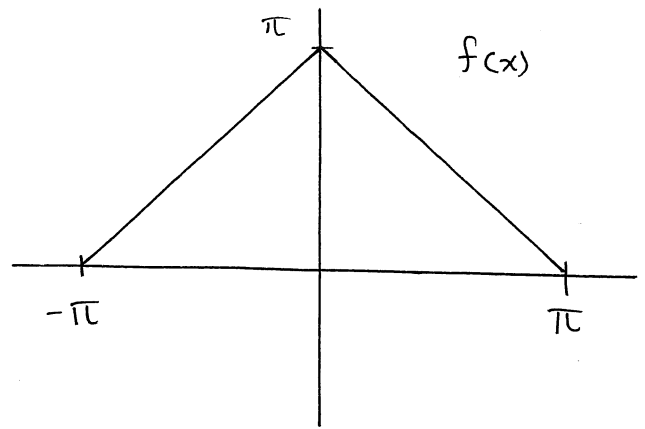


HOMEWORK 15 FOURIER SERIES

Problem 1

(i) Since $f(x)$ is even $\Rightarrow b_n = 0$ and



$f(x)$ is even

$$P = 2\pi \Rightarrow L = \pi$$

$$f(x) = \begin{cases} x + \pi & -\pi \leq x \leq 0 \\ -x + \pi & 0 \leq x \leq \pi \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} (-x + \pi) dx$$

$$= \frac{1}{\pi} \left(-\frac{x^2}{2} + \pi x \right) \Big|_0^{\pi}$$

$$= \frac{1}{\pi} \left(-\frac{\pi^2}{2} + \pi^2 \right) = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{\pi} (-x + \pi) \cos(nx) dx$$

$$= \frac{2}{\pi} \left(-\frac{x + \pi}{n} \sin(nx) + \frac{1}{n} \int_0^{\pi} \sin(nx) dx \right)$$

$$= \frac{2}{\pi} \left(-\frac{x + \pi}{n} \sin(nx) - \frac{1}{n^2} \cos(nx) \right) \Big|_0^{\pi}$$

$$= \frac{2}{\pi} \left[\left(0 - \frac{1}{n^2} \cos(n\pi) \right) - \left(0 - \frac{1}{n^2} \right) \right]$$

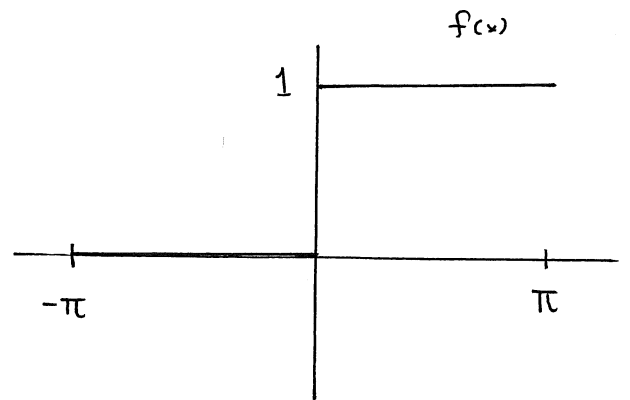
$$= \frac{2}{\pi} \left(-\frac{(-1)^n}{n^2} + \frac{1}{n^2} \right) = \frac{2}{\pi} \left(\frac{(-1)^{n+1} + 1}{n^2} \right)$$

Note: $a_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{4}{\pi n^2} & \text{if } n \text{ is odd} \end{cases}$

$$\Rightarrow f(x) = \frac{\pi}{2} + \sum_{n \text{ odd}} \frac{4}{\pi n^2} \cos(nx)$$

$$\begin{aligned} u &= -x + \pi & v &= \frac{1}{n} \sin(nx) \\ du &= -dx & dv &= \cos(nx) dx \end{aligned}$$

$$\begin{aligned}
 (ii) \quad a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \\
 &= \frac{1}{2\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx \right] \\
 &= \frac{1}{2\pi} x \Big|_0^{\pi} = \frac{1}{2\pi} (\pi) \\
 &= \frac{1}{2}
 \end{aligned}$$



$f(x)$ is neither even nor odd

$$P = 2\pi \Rightarrow L = \pi$$

$$f(x) = \begin{cases} 0 & -\pi \leq x < 0 \\ 1 & 0 \leq x \leq \pi \end{cases}$$

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} \cos(nx) dx \right] \\
 &= \frac{1}{\pi} \left(\frac{1}{n} \sin(nx) \Big|_0^{\pi} \right) = 0
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \left[\int_0^{\pi} \sin(nx) dx \right] \\
 &= \frac{1}{\pi} \left(-\frac{1}{n} \cos(nx) \right) \Big|_0^{\pi} \\
 &= \frac{1}{\pi} \left(-\frac{1}{n} \cos(n\pi) + \frac{1}{n} \cos(0) \right) \\
 &= \frac{1}{\pi} \left(-\frac{1}{n} (-1)^n + \frac{1}{n} \right) = \frac{(-1)^{n+1} + 1}{\pi n}
 \end{aligned}$$

Note: $b_n = \begin{cases} 0 & n \text{ even} \\ \frac{2}{\pi n} & n \text{ odd} \end{cases}$

$$\Rightarrow f(x) = \frac{1}{2} + \sum_{n \text{ odd}} \frac{2}{\pi n} \sin(nx)$$

(iii) Since f is odd $\Rightarrow a_0 = a_n = 0$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx$$

$$= \frac{2}{\pi} \left[\int_0^{\pi/2} x \sin(nx) dx + \int_{\pi/2}^{\pi} \frac{\pi}{2} \sin(nx) dx \right]$$

$$= \frac{2}{\pi} [\textcircled{1} + \textcircled{2}]$$

$$\textcircled{1} = -\frac{x}{n} \cos(nx) + \frac{1}{n} \int \cos(nx) dx \Big|_0^{\pi/2} \quad \begin{array}{l} u=x \\ du=dx \end{array} \quad \begin{array}{l} v = -\frac{1}{n} \cos(nx) \\ dv = \sin(nx) dx \end{array}$$

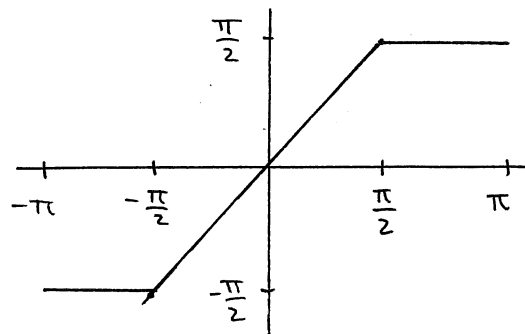
$$= -\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \Big|_0^{\pi/2}$$

$$= -\frac{\pi}{2n} \cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\textcircled{2} = -\frac{\pi}{2n} \cos(nx) \Big|_{\pi/2}^{\pi} = -\frac{\pi}{2n} \cos(n\pi) + \frac{\pi}{2n} \cos\left(\frac{n\pi}{2}\right) = -\frac{\pi}{2n} (-1)^n + \frac{\pi}{2n} \cos\left(\frac{n\pi}{2}\right)$$

$$\frac{2}{\pi} [\textcircled{1} + \textcircled{2}] = \frac{2}{\pi} \left[\frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right) - \frac{\pi}{2n} (-1)^n \right]$$

$$\Rightarrow f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right) - \frac{\pi}{2n} (-1)^n \right] \sin(nx)$$



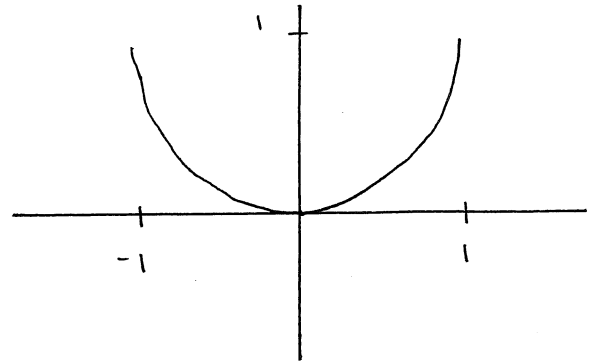
$f(x)$ is odd

$$P = 2\pi \Rightarrow L = \pi$$

$$f(x) = \begin{cases} -\frac{\pi}{2} & -\pi \leq x \leq -\frac{\pi}{2} \\ x & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ \frac{\pi}{2} & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

Problem 2

Since f is even $\rightarrow b_n = 0$



$f(x)$ is even

$$P = 2 \Rightarrow L = 1$$

$$a_0 = \frac{1}{1} \int_0^1 x^2 dx$$

$$= \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$

$$a_n = \frac{2}{1} \int_0^1 x^2 \cos(n\pi x) dx$$

$$= 2 \left(\frac{x^2}{n\pi} \sin(n\pi x) - \frac{2}{n\pi} \int x \sin(n\pi x) dx \right) \Big|_0^1$$

$$u = x^2 \quad v = \frac{1}{n\pi} \sin(n\pi x)$$

$$du = 2x dx \quad dv = \cos(n\pi x) dx$$

$$= 2 \left(\frac{x^2}{n\pi} \sin(n\pi x) - \frac{2}{n\pi} \left(\frac{-x}{n\pi} \cos(n\pi x) + \frac{1}{n\pi} \int \cos(n\pi x) dx \right) \right) \Big|_0^1$$

$$u = x \quad v = -\frac{1}{n\pi} \cos(n\pi x)$$

$$du = dx \quad dv = \sin(n\pi x) dx$$

$$= 2 \left[\frac{x^2}{n\pi} \sin(n\pi x) + \frac{2x}{n^2\pi^2} \cos(n\pi x) - \frac{2}{n^2\pi^2} \left(\frac{1}{n\pi} \sin(n\pi x) \right) \right] \Big|_0^1$$

$$= 2 \left[\left(\frac{1}{n\pi} \sin(n\pi) + \frac{2}{n^2\pi^2} \cos(n\pi) - \frac{2}{n^3\pi^3} \sin(n\pi) \right) - \left(0 + 0 - \frac{2}{n^3\pi^3} \sin(0) \right) \right]$$

$$= 2 \left(0 + \frac{2}{n^2\pi^2} (-1)^n - 0 + 0 \right)$$

$$= \frac{4}{n^2\pi^2} (-1)^n$$

$$\Rightarrow f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi x)$$

Extra Credit

Since $f(x) = x^2$ $-1 \leq x \leq 1$ and

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi x)$$

$$\Rightarrow x^2 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi x) \quad \text{for all } x \text{ in } [-1, 1]$$

let $x=1$

$$\begin{aligned} \Rightarrow 1 &= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi) \\ &= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} (-1)^n \\ &= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{((-1)^n)^2}{n^2} \\ &= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \end{aligned}$$

Solving for $\sum_{n=1}^{\infty} \frac{1}{n^2}$ we have :

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$